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Machine Learning Guided Adaptive Scheduling for Multi-core Real-time Operating Systems

Florianópolis
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Certificamos que esta é a **versão original e final** do trabalho de conclusão que foi julgado adequado para obtenção do título de Mestre em Ciência da Computação.

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ABSTRACT

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DVFS	Dynamic voltage and frequency scaling
ML	Machine Learning
OS	Operating System
PMC	Performance Monitoring Counter
RTOS	Real-time Operating Systems

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1 INTRODUCTION

Currently, several areas of technology require software that must execute in real-time. This often includes autonomous vehicles, aviation, robotics, industrial systems, and other critical systems where the software is directly responsible for system safety and control. In these domains, a failure to meet timing constraints during execution can lead to severe consequences. Thus, to ensure predictable execution and meet these timing requirements, Real-time Operating Systemss (RTOSs) are utilized.

These systems are commonly employed in embedded scenarios, where other variables must also be considered, such as energy consumption and thermal limits (Rahim; Nair; Devaraj, 2025). Dynamic voltage and frequency scaling (DVFS) can be used to reduce the energy consumption, but, to be used in a RTOS, there must be a careful consideration for the timing constraints and how different DVFS levels can alter the utilization of tasks. An algorithm for task utilization prediction could be employed to address this concern, enabling the safer usage of DVFS, Since the utilization observed in the current DVFS level does not transfer directly to other levels, it must be predicted before any level change. Moreover, task utilization prediction can be useful for other applications, such as load balancing, where an estimation of utilization could be made for different task configurations.

However, predicting the utilization is a considerable challenge in modern multicore systems (Islam *et al.*, 2018; Hoffmann; Fröhlich, 2021). One of the major challenges introduced by multicore processors is interference, which arises from task contention for shared resources such as main memory, caches, and I/O peripherals. This contention introduces non-deterministic delays into task execution, making it difficult to accurately estimate the execution times and the overall utilization (Gracioli *et al.*, 2019; Horstmann; Hoffmann; Fröhlich, 2019; Aceituno *et al.*, 2023; Lugo *et al.*, 2022) with the usage of simple analytical methods. The problem is aggravated in heterogeneous multicore processors, which display cores with different characteristics (Mrabet; Karamti; Mahfoudhi, 2024; Jadon *et al.*, 2024).

In recent years, several works have proposed the use of Performance Monitoring Counters (PMCs) in predictive algorithms, relying on hardware information to guide decisions. Most of these approaches rely on Machine Learning (ML) techniques, however, they also report difficulties and pitfalls that hinder their adoption. ML solutions usually introduce prohibitive overhead and fail to generalize well to new task-sets and platforms (Rahim; Nair; Devaraj, 2025). Moreover, the use of PMCs makes it difficult to design portable predictive algorithms. Although some works propose solutions to this issue, they remain limited by problems such as the divergence of PMC semantics across architectures, lack of counters in some platform implementations and also noise introduced by the Operating System (OS) and other external sources (Das *et al.*, 2019).

Considering this scenario, it would be useful to address the main problems associated with the use of predictive algorithms in RTOS, specifically with the usage of PMC enabled ML based algorithms. This dissertation focuses on these problems and offers an analysis based on data collected from a RTOS running on multicore RISC-V and ARM platforms, with an additional proposed solution that improves upon the current approaches and offers a portable ML guided, PMC enabled, multicore capable, predictive algorithm applied to energy consumption optimization and load balancing. The proposed solution employs lightweight task-local ML models executed in idle time for minimal interference with the tasks.

1.1 OBJECTIVES

The sections below present the general and specific objectives.

1.1.1 General Objective

The objective of this work is to improve upon the current approaches to task utilization prediction by developing a portable algorithm for utilization prediction in multicore systems with the usage of ML models and PMC. The main focus will be in the usage of this algorithm for energy optimization and load balancing.

1.1.2 Specific Objectives

- Provide an evaluation of common problems in statistical approaches to utilization prediction in multicore RISC-V platforms, evaluating the performance of the approach of Hoffmann and Fröhlich (2021);
- Provide an implementation of a ML guided portable algorithm for utilization prediction in multicore systems with the use of PMC, with the focus in energy optimization and load balancing;
- Perform an evaluation of the implementation in a RISC-V platform;
- Perform an evaluation of the implementation in an ARM platform.

2 THEORETICAL BASIS

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3 CONCLUSION

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APPENDIX A – APPENDIX

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